Separation of variables

At the very beginning of this course, we talked about the difference between ordinary differential equations (ODEs) and partial differential equations (PDEs).

Remember that, put simply, ordinary differential equations are equations defined with ordinary derivatives, while partial differential equations are equations defined with partial derivatives. Here are some examples.

ODEs	**PDEs**
$8y''' + 2y' + \cos y = e^x$	$\dfrac{\partial u}{\partial t} = k\dfrac{\partial^2 u}{\partial x^2}$
$2y'' + 5y' = xy$	$\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$

Partial derivatives

What we see from these examples is that ordinary derivatives are the derivatives of single variable functions. For instance, given a function $y(x)$ that's defined for y in terms of x, its derivatives will always be with respect to the independent variable, x. Its first derivative $y'(x)$ is a derivative with respect to x, its second derivative $y''(x)$ is a derivative with respect to x, etc.

But partial derivatives are the derivatives of multivariable functions. For instance, a function $f(x, y)$ defined for f in terms of x and y can have derivatives with respect to one or both of its independent variables.

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It has two first derivatives,

$\frac{\partial f}{\partial x}$ the partial derivative with respect to x

$\frac{\partial f}{\partial y}$ the partial derivative with respect to y

and four second derivatives,

$\frac{\partial^2 f}{\partial x^2}$ the partial derivative with respect to x, then x again

$\frac{\partial^2 f}{\partial y^2}$ the partial derivative with respect to y, then y again

$\frac{\partial^2 f}{\partial y \partial x}$ the partial derivative with respect to x, then y

$\frac{\partial^2 f}{\partial x \partial y}$ the partial derivative with respect to y, then x

If the function f is well-behaved enough, these last two second derivatives, the **mixed derivatives**, will be equivalent, which means it won't make any difference whether we differentiate first with respect to x and then y, or y and then x.

Hopefully this is all just review, since partial derivatives are always covered as part of any multivariable calculus course.

But with this brief review out of the way, we want to start looking at solving partial differential equations. So far, we've been focusing entirely on ODEs, but now we'll turn toward PDEs. In fact, the main reason we

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learned to find Fourier series representations was to prepare us to solve some partial differential equations.

Solving by separating variables

Much earlier, we learned how to use separation of variables to solve first order ordinary differential equations. But we can also use separation of variables to solve partial differential equations.

To solve a linear homogeneous partial differential equation in x and t by separating variables, we assume that the PDE has a **product solution** in the form

$$u(x,t) = v(x)w(t)$$

We use this assumption for the solution equation because it'll allow us to simplify the partial differential equation to multiple ordinary differential equations, and of course we're already really comfortable solving ODEs.

Let's do an example to see how to use this product solution.

Example

Use the product solution to separate variables and reduce the partial differential equation into a pair of ordinary differential equations.

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

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We'll find the first and second derivatives of the product solution.

$$u(x,t) = v(x)w(t)$$

$$\frac{\partial u}{\partial t} = v(x)w'(t)$$

$$\frac{\partial u}{\partial x} = v'(x)w(t) \qquad \frac{\partial^2 u}{\partial x^2} = v''(x)w(t)$$

Then we'll plug the derivatives into the partial differential equation.

$$v(x)w'(t) = kv''(x)w(t)$$

If we rewrite the equation in Leibniz notation, it becomes more obvious that the equation is now entirely in terms of ordinary derivatives. All partial derivatives have been substituted out.

$$v(x)\frac{dw}{dt} = k\frac{d^2v}{dx^2}w(t)$$

Now we can separate variables.

$$\left(\frac{1}{kw(t)}\right)\frac{dw}{dt} = \left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2}$$

The left side of this equation is a function entirely in terms of the independent variable t, while the right side of the equation is a function entirely in terms of the independent variable x.

Which means the equation is telling us that a function in terms of t will always be equivalent to a function in terms of x, regardless of the values of x and t that we choose. But the only way this can possibly be true is if both

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functions are constant. In fact, they both have to be equal to the same constant.

We don't yet know the value of that constant, but let's call it $-\lambda$. This $-\lambda$ value is the **separation constant**.

$$\left(\frac{1}{kw(t)}\right)\frac{dw}{dt} = \left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

Now that we have a three-part equation, we can break it apart into two separate equations.

$$\left(\frac{1}{kw(t)}\right)\frac{dw}{dt} = -\lambda$$

$$\frac{dw}{dt} = -k\lambda w(t)$$

and

$$\left(\frac{1}{v(x)}\right)\frac{d^2v}{dx^2} = -\lambda$$

$$\frac{d^2v}{dx^2} = -\lambda v(x)$$

Using the product solution in this last example allowed us to change the partial differential equation into a pair of ordinary differential equations.

We'll talk later about how to solve this pair of ODEs, but for now we just want to understand the idea of separation of variables and the use of the

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product solution, since we'll be using variations of this method to solve all of the partial differential equations that we'll be looking at over the next few lessons.

That being said, now we want to turn our attention toward boundary value problems and the boundary conditions we use with this product solution and separation of variables.

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